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RESEARCH PAPER

# Full Dynamic Modeling of the General Stewart Platform Manipulator via Kane's Method

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**Abstract** Full dynamic modeling of the general Stewart platform manipulator is a complex task due to its closed-loop structure. In this paper, a complete model of inverse dynamics of the most general Stewart platform manipulator, without any simplification on dynamic properties of its components, is presented. It is noted that the equations obtained considering the comprehensive and detailed description of the Stewart system bear a computational burden considerably less than that from the more traditional methods of dynamic modeling, and this was indeed verified in the results obtained through simulations performed using the algorithm presented in this research.

**Keywords** General Stewart platform · Kane's method · Inverse dynamics · Computational cost · Friction

## 1 Introduction

Stewart platform manipulator consists of a fixed platform (base) and a moving platform connected together by six identical legs. It has six degrees of freedom and has several advantages over conventional serial manipulators, like more stiffness, accuracy, and more power to weight ratio.

Generally, dynamic modeling of a parallel manipulator is complicated due to their closed-loop nature. Simulating these manipulators is, however, an important step in the design of a proper controller. Several approaches are used to analyze the dynamics of the Stewart platform manipulator. Dasgupta and Mruthyunjaya (1998) and Guo and Li (2006) use the Newton–Euler method for the Stewart system, while Khalil and Ibrahim (2007) use this method to present an algorithm for the dynamic modeling of general parallel manipulators. Lagrange method was also used by some researchers, as Nguyen and Pooran (1989) and Di Gregorio and Parenti-Castelli (2004). Tsai (2000) uses the principle of virtual work and presents a fast algorithm for calculating the inverse dynamics of the Stewart platform manipulator. Lopes (2008) also presented an algorithm using the generalized momentum approach. Some researchers also combined such methods as the recursive matrix (Staicu 2011), energy equivalence (Abdellatif and Heimann 2009), and screw theory (Amirouche 2007) with classical approaches to present more efficient algorithms in analyzing dynamics of this manipulator. These models apply different levels of simplification on the platform dynamics and geometric properties, such as point masses assumption for the legs, with the center of masses of legs on their central axes, neglecting product of inertias of legs, and with the assumption of no friction in the different joints of the manipulator. Among the approaches mentioned, the work conducted by Dasgupta and Mruthyunjaya (1998) is the most general approach, but only assumes viscous friction in the joints and the algorithm is not so time efficient. The work conducted by Tsai (2000) is very fast but does not include friction in the joints, with many simplifications assumed for the legs. Our goal is using Kane's method to present a time-efficient algorithm for solving the inverse dynamics of the general Stewart platform manipulator without any simplification made on its properties. In our model, we consider a real mass model of legs and implement a complete continuous friction model (suitable for control

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applications) for actuators by introducing an isolated term in the final equations. This can also be easily replaced by other friction models without any further modification on the other terms. Kane's method for multi-body systems is used to achieve a considerably low computational cost, and the Jacobian matrices from (Tsai 2000) are also used to put the final equations of motion in compact form.

## 2 Kinematics

The Stewart platform manipulator has two kinds of structure, a 6-SPS[1] structure has several passive degrees of freedom (legs can rotate freely about their axes) and may suffer from complications in applications, so we consider the 6-UPS[2] structure in the analysis of the system. The universal joint located on the fixed platform is denoted as $B_i$, and the spherical joints are named as $P_i$. We attach a fixed frame $B(x, y, z)$ to the base and a moving frame $P(u, v, w)$ to the center of mass of the moving platform (Fig. 1).

We can write the following equations for the kinematics of each leg,

$$\boldsymbol{s_i} = \left(\boldsymbol{t} + {}^{B}\boldsymbol{R_P}\boldsymbol{p}'_i - \boldsymbol{b_i}\right)/d_i \tag{1}$$

$$d_i = \left\|\boldsymbol{t} + {}^{B}\boldsymbol{R_P}\boldsymbol{p}'_i - \boldsymbol{b_i}\right\| \tag{2}$$

where $\boldsymbol{t} = [x \quad y \quad z]^{\mathrm{T}}$ is the position of the moving frame with respect to the base frame. Here, the rotation matrix of the moving frame is defined as follows. A rotation $\varphi_x$ about the $x$ axis of the fixed frame followed by a rotation $\varphi_y$ about the $y$ axis of the fixed frame, and a rotation $\varphi_z$ about the $z$ axis of the fixed frame

$${}^{B}\boldsymbol{R_P} = \begin{bmatrix} c\varphi_z c\varphi_y & c\varphi_z s\varphi_y s\varphi_x - s\varphi_z c\varphi_x & c\varphi_z s\varphi_y c\varphi_x + s\varphi_z s\varphi_x \\ s\varphi_z c\varphi_y & s\varphi_z s\varphi_y s\varphi_x + c\varphi_z c\varphi_x & s\varphi_z s\varphi_y c\varphi_x - c\varphi_z s\varphi_x \\ -s\varphi_y & c\varphi_y s\varphi_x & c\varphi_y c\varphi_x \end{bmatrix} \tag{3}$$

where 's' and 'c' are abbreviations of sin and cos trigonometric functions. For the future analysis, we also attach local frames $C_i(x_i, y_i, z_i)$ to each leg, with the origin of each frame located at the universal joint of the leg. The rotation of these frames is represented by leg rotation matrices ${}^{B}\boldsymbol{R_i}$ as follows. A rotation $\varphi_i$ about the $z_i^{'}$ axis of the intermediate local frame is followed by a rotation $\theta_i$ about the $y_i$ axis of the leg local frame. The leg rotation matrices can then be written as the following:

$${}^{B}\boldsymbol{R_i} = \begin{bmatrix} c\varphi_i c\theta_i & -s\varphi_i & c\varphi_i s\theta_i \\ s\varphi_i c\theta_i & c\varphi_i & s\varphi_i s\theta_i \\ -s\theta_i & 0 & c\theta_i \end{bmatrix} \tag{4}$$

[1] 6- Spherical, Prismatic, Spherical.

[2] 6- Universal, Prismatic, Spherical.

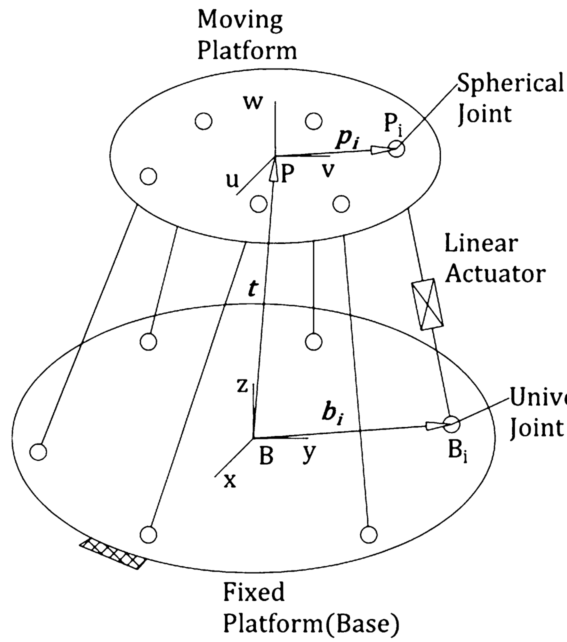


**Fig. 1** Schematic representation of Stewart platform

Since $\boldsymbol{s_i}$ is the unit vector along the $z_i$ axis, it is equal to the third column of the leg rotation matrix, i.e.,

$$\boldsymbol{s_i} = [c\varphi_i s\theta_i \quad s\varphi_i s\theta_i \quad c\theta_i]^{\mathrm{T}} \tag{5}$$

By equating Eq. (5) to values calculated from Eq. (1), all the trigonometric functions for the leg rotation matrix are obtained as

$$\begin{cases} c\theta_i = s_{iz} \\ s\theta_i = \sqrt{s_{ix}^2 + s_{iy}^2} \\ s\varphi_i = \dfrac{s_{iy}}{s\theta_i} \\ c\varphi_i = \dfrac{s_{ix}}{s\theta_i} \end{cases} ,0 \le \theta \le \pi \tag{6}$$

where $s_{ix}$, $s_{iy}$, and $s_{iz}$ are the $x$, $y$, and $z$ components of $\boldsymbol{s_i}$ (Fig. 2).

## 3 Velocity and Accelerations

### 3.1 Velocity Analysis

The generalized speed vector is defined as the following

$$\dot{\boldsymbol{x}}_{\boldsymbol{p}} = \left[\dot{\boldsymbol{t}}^T \quad \boldsymbol{\omega}_{\boldsymbol{p}}^T\right]^{\mathrm{T}} \tag{7}$$

where $\dot{\boldsymbol{t}}$ and $\boldsymbol{\omega_p}$ are the linear and angular velocities of the platform. The linear velocity of point $P_i$ expressed in fixed frame can readily be calculated as

$$\boldsymbol{v_{pi}} = \boldsymbol{\omega_p} \times \boldsymbol{p_i} + \dot{\boldsymbol{t}} \tag{8}$$

where $\boldsymbol{p_i}$ is defined as $\boldsymbol{p_i} = {}^{B}\boldsymbol{R_P}\boldsymbol{p}_i^{'}$.

The above velocity can be expressed in the local frame as

$$ {}^{i}\boldsymbol{v}_{p_i} = {}^{i}\boldsymbol{R}_{\boldsymbol{B}}\boldsymbol{v}_{p_i}, \quad {}^{i}\boldsymbol{R}_{\boldsymbol{B}} = {}^{B}\boldsymbol{R}_{i}^{T} \tag{9} $$

This velocity also can be obtained as the following

$$ {}^{i}\boldsymbol{v}_{p_i} = d_i {}^{i}\boldsymbol{\omega}_{\boldsymbol{i}} \times {}^{i}\boldsymbol{s}_{\boldsymbol{i}} + \dot{d}_i {}^{i}\boldsymbol{s}_{\boldsymbol{i}} \tag{10} $$

where ${}^{i}\boldsymbol{s}_{\boldsymbol{i}}$ is equal to $[0 \quad 0 \quad 1]^{\mathrm{T}}$ and $\dot{d}_i$ is the linear velocity between two sections of the legs relative to each other, and ${}^{i}\boldsymbol{\omega}_{\boldsymbol{i}}$ is the angular velocity vector of the legs expressed in the local frame.

Since the legs do not rotate about their longitudinal axis, we have ${}^{i}\boldsymbol{\omega}_{\boldsymbol{i}}^{\mathrm{T}}\boldsymbol{s}_{\boldsymbol{i}} = 0$. Dot-multiplying Eq. (10) with ${}^{i}\boldsymbol{s}_{\boldsymbol{i}}$ will result in the following expression for $\dot{d}_i$,

$$ \dot{d}_i = {}^{i}v_{p_{iz}} \tag{11} $$

where ${}^{i}v_{p_{iz}}$ is the third component of ${}^{i}\boldsymbol{v}_{p_i}$. By cross-multiplying Eq. (10) with ${}^{i}\boldsymbol{s}_{\boldsymbol{i}}$, the angular velocity of the legs can be obtained as

$$ {}^{i}\boldsymbol{\omega}_{\boldsymbol{i}} = \frac{1}{d_i}\left({}^{i}\boldsymbol{s}_{\boldsymbol{i}} \times {}^{i}\boldsymbol{v}_{p_i}\right) = \frac{1}{d_i}\begin{bmatrix} -{}^{i}v_{p_{iy}} \\ {}^{i}v_{p_{ix}} \\ 0 \end{bmatrix} \tag{12} $$

By knowing $\dot{d}_i$ and ${}^{i}\boldsymbol{\omega}_{\boldsymbol{i}}$, we can calculate the velocities of each of the two sections of the leg as

$$ {}^{i}\boldsymbol{v}_{1i} = {}^{i}\boldsymbol{\omega}_{\boldsymbol{i}} \times \boldsymbol{e}_{\boldsymbol{i}} = \frac{1}{d_i}\begin{bmatrix} {}^{i}v_{p_{ix}}e_{iz} \\ {}^{i}v_{p_{iy}}e_{iz} \\ -{}^{i}v_{p_{ix}}e_{ix} - {}^{i}v_{p_{iy}}e_{iy} \end{bmatrix} \tag{13} $$

$$ \begin{aligned} {}^{i}\boldsymbol{v}_{2i} &= {}^{i}\boldsymbol{\omega}_{\boldsymbol{i}} \times \left(d_i {}^{i}\boldsymbol{s}_{\boldsymbol{i}} + \boldsymbol{h}_{\boldsymbol{i}}\right) + \dot{d}_i {}^{i}\boldsymbol{s}_{\boldsymbol{i}} \\ &= \frac{1}{d_i}\begin{bmatrix} {}^{i}v_{p_{ix}}(h_{iz} + d_i) \\ {}^{i}v_{p_{iy}}(h_{iz} + d_i) \\ -{}^{i}v_{p_{ix}}h_{ix} - {}^{i}v_{p_{iy}}h_{iy} + d_i {}^{i}v_{p_{iz}} \end{bmatrix} \end{aligned} \tag{14} $$

where ${}^{i}\boldsymbol{v}_{1i}$ and ${}^{i}\boldsymbol{v}_{2i}$ are the absolute linear velocities of the leg links center of masses expressed in leg local frames.

${}^{i}\boldsymbol{\omega}_{\boldsymbol{p}}$ which is the angular velocity of the platform stated in the $i$th leg frame is also introduced for further use in obtaining friction in spherical joints as the following

$$ {}^{i}\boldsymbol{\omega}_{\boldsymbol{p}} = {}^{i}\boldsymbol{R}_{\boldsymbol{B}}\boldsymbol{\omega}_{\boldsymbol{p}} \tag{15} $$

## 3.2 Acceleration Analysis

Taking the time-derivative of Eq. (8) yields the acceleration of point $P_i$ expressed in fixed frame.

$$ \dot{\boldsymbol{v}}_{p_i} = +\dot{\boldsymbol{\omega}}_{\boldsymbol{p}} \times \boldsymbol{p}_{\boldsymbol{i}} + \boldsymbol{\omega}_{\boldsymbol{p}} \times \left(\boldsymbol{\omega}_{\boldsymbol{p}} \times \boldsymbol{p}_{\boldsymbol{i}}\right) \tag{16} $$

where  and $\dot{\boldsymbol{\omega}}_{\boldsymbol{p}}$ are the linear and angular accelerations of the moving platform. Expressing these accelerations in the legs local frame will result in

$$ {}^{i}\dot{\boldsymbol{v}}_{p_i} = {}^{i}\boldsymbol{R}_{\boldsymbol{B}}\dot{\boldsymbol{v}}_{p_i} \tag{17} $$

On the other hand, acceleration of point $P_i$ expressed in the legs local frame can be obtained by taking the time-derivative of Eq. (10) as

$$ \begin{aligned} {}^{i}\dot{\boldsymbol{v}}_{p_i} = {}& \ddot{d}_i {}^{i}\boldsymbol{s}_{\boldsymbol{i}} + d_i {}^{i}\dot{\boldsymbol{\omega}}_{\boldsymbol{i}} \times {}^{i}\boldsymbol{s}_{\boldsymbol{i}} + d_i {}^{i}\boldsymbol{\omega}_{\boldsymbol{i}} \times \left({}^{i}\boldsymbol{\omega}_{\boldsymbol{i}} \times {}^{i}\boldsymbol{s}_{\boldsymbol{i}}\right) + 2\dot{d}_i {}^{i}\boldsymbol{\omega}_{\boldsymbol{i}} \\ & \times {}^{i}\boldsymbol{s}_{\boldsymbol{i}} \end{aligned} \tag{18} $$

where $\ddot{d}_i$ is the relative acceleration between the two sections of the legs relative to each other obtained by dot-multiplying Eq. (18) with ${}^{i}\boldsymbol{s}_{\boldsymbol{i}}$.

$$ \ddot{d}_i = {}^{i}\dot{v}_{p_{iz}} - d_i {}^{i}\boldsymbol{\omega}_{\boldsymbol{i}}^{2} = {}^{i}\dot{v}_{p_{iz}} + \frac{\left({}^{i}v_{p_{ix}}^{2} + {}^{i}v_{p_{iy}}^{2}\right)}{d_i} \tag{19} $$

To obtain the angular acceleration of the legs, one must cross-multiply Eq. (18) with ${}^{i}\boldsymbol{s}_{\boldsymbol{i}}$. Hence,

$$ {}^{i}\dot{\boldsymbol{\omega}}_{\boldsymbol{i}} = \frac{{}^{i}\boldsymbol{s}_{\boldsymbol{i}} \times {}^{i}\dot{\boldsymbol{v}}_{p_i} - 2\dot{d}_i {}^{i}\boldsymbol{\omega}_{\boldsymbol{i}}}{d_i} = \frac{1}{d_i^2}\begin{bmatrix} -d_i {}^{i}\dot{v}_{p_{iy}} + 2 {}^{i}v_{p_{iz}} {}^{i}v_{p_{iy}} \\ d_i {}^{i}\dot{v}_{p_{ix}} - 2 {}^{i}v_{p_{iz}} {}^{i}v_{p_{ix}} \\ 0 \end{bmatrix} \tag{20} $$

The acceleration vector of the leg links center of masses expressed in leg local frames can be calculated as

$$ {}^{i}\dot{\boldsymbol{v}}_{1i} = {}^{i}\dot{\boldsymbol{\omega}}_{\boldsymbol{i}} \times \boldsymbol{e}_{\boldsymbol{i}} + {}^{i}\boldsymbol{\omega}_{\boldsymbol{i}} \times \left({}^{i}\boldsymbol{\omega}_{\boldsymbol{i}} \times \boldsymbol{e}_{\boldsymbol{i}}\right) \tag{21} $$

$$ \begin{aligned} {}^{i}\dot{\boldsymbol{v}}_{2i} = {}& \ddot{d}_i {}^{i}\boldsymbol{s}_{\boldsymbol{i}} + {}^{i}\dot{\boldsymbol{\omega}}_{\boldsymbol{i}} \times \left(d_i {}^{i}\boldsymbol{s}_{\boldsymbol{i}} + \boldsymbol{h}_{\boldsymbol{i}}\right) + {}^{i}\boldsymbol{\omega}_{\boldsymbol{i}} \\ & \times \left({}^{i}\boldsymbol{\omega}_{\boldsymbol{i}} \times \left(d_i {}^{i}\boldsymbol{s}_{\boldsymbol{i}} + \boldsymbol{h}_{\boldsymbol{i}}\right)\right) + 2\dot{d}_i {}^{i}\boldsymbol{\omega}_{\boldsymbol{i}} \times {}^{i}\boldsymbol{s}_{\boldsymbol{i}} \end{aligned} \tag{22} $$

Simplifying Eqs. (21) and (22) yields

$$ {}^{\mathbf{i}}\dot{v}_{\mathbf{1i}} = \frac{1}{d_i^2}\begin{bmatrix} \left(d_i^i\dot{v}_{p_{ix}} - 2^i v_{p_{iz}}^i v_{p_{ix}}\right)e_{iz} - d_i^{2i}v_{p_{ix}}({}^i v_{p_{ix}}e_{ix} + {}^i v_{p_{iy}}e_{iy}) \\ \left(d_i^i\dot{v}_{p_{iy}} - 2^i v_{p_{iz}}^i v_{p_{iy}}\right)e_{iz} - d_i^{2i}v_{p_{iy}}({}^i v_{p_{ix}}e_{ix} + {}^i v_{p_{iy}}e_{iy}) \\ -\left(d_i^i\dot{v}_{p_{ix}} - 2^i v_{p_{iz}}^i v_{p_{ix}}\right)e_{ix} - (d_i^i\dot{v}_{p_{iy}} - 2^i v_{p_{iz}}^i v_{p_{iy}})e_{iy} - ({}^i v_{p_{ix}}^2 + {}^i v_{p_{iy}}^2)e_{iz} \end{bmatrix} \tag{23} $$

$$ {}^{\mathbf{i}}\dot{v}_{\mathbf{2i}} = \frac{1}{d_i^2}\begin{bmatrix} d_i(h_{iz} + d_i)^i\dot{v}_{p_{ix}} - 2h_{iz}^i v_{p_{iz}}^i v_{p_{ix}} - d_i^{2i}v_{p_{ix}}({}^i v_{p_{ix}}h_{ix} + {}^i v_{p_{iy}}h_{iy}) \\ d_i(h_{iz} + d_i)^i\dot{v}_{p_{iy}} - 2h_{iz}^i v_{p_{iz}}^i v_{p_{iy}} - d_i^{2i}v_{p_{iy}}({}^i v_{p_{ix}}h_{ix} + {}^i v_{p_{iy}}h_{iy}) \\ -(d_i^i\dot{v}_{p_{ix}} - 2^i v_{p_{iz}}^i v_{p_{ix}})h_{ix} - (d_i^i\dot{v}_{p_{iy}} - 2^i v_{p_{iz}}^i v_{p_{iy}})h_{iy} + d_i^{2i}\dot{v}_{p_{iz}} - ({}^i v_{p_{ix}}^2 + {}^i v_{p_{iy}}^2)h_{iz} \end{bmatrix} \tag{24} $$

## 3.3 Jacobians

The Jacobian matrices are required in order to write the equation of motion of the platform in compact form, and we use the same procedure as (Tsai 2000). Starting from Eq. (8), it can be rewritten in matrix form as

$$ \boldsymbol{v}_{p_i} = \boldsymbol{J}_{p_i}\dot{\boldsymbol{x}}_{\boldsymbol{p}} \tag{25} $$

where $\boldsymbol{J}_{p_i}$ is defined as

$$ \boldsymbol{J}_{p_i} = \begin{bmatrix} 1 & 0 & 0 & 0 & p_{iz} & -p_{iy} \\ 0 & 1 & 0 & -p_{iz} & 0 & p_{ix} \\ 0 & 1 & 0 & p_{iy} & -p_{ix} & 0 \end{bmatrix} \tag{26} $$

Also, Eq. (9) can be written as

$$ {}^{i}\boldsymbol{v}_{p_i} = {}^{i}\boldsymbol{J}_{p_i}\dot{\boldsymbol{x}}_{\boldsymbol{p}} \tag{27} $$

where

$$ {}^{i}\boldsymbol{J}_{p_i} = {}^{i}\boldsymbol{R}_{\boldsymbol{B}}\boldsymbol{J}_{p_i} = \begin{bmatrix} {}^{i}\boldsymbol{J}_{p_{ix}} \\ {}^{i}\boldsymbol{J}_{p_{iy}} \\ {}^{i}\boldsymbol{J}_{p_{iz}} \end{bmatrix} \tag{28} $$

and ${}^{i}\boldsymbol{J}_{p_{ix}}$, ${}^{i}\boldsymbol{J}_{p_{iy}}$, and ${}^{i}\boldsymbol{J}_{p_{iz}}$ are the rows of ${}^{i}\boldsymbol{J}_{p_i}$.

Following same approach, Eq. (11) is rewritten as

$$ \dot{d}_i = {}^{i}\boldsymbol{J}_{p_{iz}}\dot{\boldsymbol{x}}_{\boldsymbol{p}} \tag{29} $$

The above equation in matrix form can be written as

$$ \dot{\boldsymbol{q}} = \boldsymbol{J}_{\boldsymbol{q}}\dot{\boldsymbol{x}}_{\boldsymbol{p}} \tag{30} $$

where $\boldsymbol{J}_{\boldsymbol{q}}$ is the Jacobian matrix for the manipulator, and $\dot{\boldsymbol{q}} = \begin{bmatrix} \dot{d}_1 & \cdots & \dot{d}_6 \end{bmatrix}^{\mathrm{T}}$. This matrix is written as

$$ \boldsymbol{J}_{\boldsymbol{q}} = \begin{bmatrix} {}^{1}\boldsymbol{J}_{p_{1z}} \\ {}^{2}\boldsymbol{J}_{p_{2z}} \\ {}^{3}\boldsymbol{J}_{p_{3z}} \\ {}^{4}\boldsymbol{J}_{p_{4z}} \\ {}^{5}\boldsymbol{J}_{p_{5z}} \\ {}^{6}\boldsymbol{J}_{p_{6z}} \end{bmatrix} \tag{31} $$

In a similar manner, Eqs. (12), (13), and (14) are written as

$$ {}^{i}\boldsymbol{\omega}_{\boldsymbol{i}} = \boldsymbol{J}_{{}^{i}\omega_i}\dot{\boldsymbol{x}}_{\boldsymbol{p}} \tag{32} $$

$$ {}^{i}\boldsymbol{v}_{1i} = \boldsymbol{J}_{v_{1i}}\dot{\boldsymbol{x}}_{\boldsymbol{p}} \tag{33} $$

$$ {}^{i}\boldsymbol{v}_{2i} = \boldsymbol{J}_{v_{2i}}\dot{\boldsymbol{x}}_{\boldsymbol{p}} \tag{34} $$

where $\boldsymbol{J}_{{}^{i}\omega_i}$ is the Jacobian matrix for the leg angular velocities, written as

$$ \boldsymbol{J}_{{}^{i}\omega_i} = \frac{1}{d_i}\begin{bmatrix} -{}^{i}\boldsymbol{J}_{p_{iy}} \\ {}^{i}\boldsymbol{J}_{p_{ix}} \\ 0_{1\times 6} \end{bmatrix} \tag{35} $$

$\boldsymbol{J}_{v_{1i}}$ and $\boldsymbol{J}_{v_{2i}}$ are the Jacobian matrices for the leg links linear velocities, which are written in the form

$$ \boldsymbol{J}_{v_{1i}} = \frac{1}{d_i}\begin{bmatrix} {}^{i}\boldsymbol{J}_{p_{ix}} e_{iz} \\ {}^{i}\boldsymbol{J}_{p_{iy}} e_{iz} \\ -{}^{i}\boldsymbol{J}_{p_{ix}} e_{ix} - {}^{i}\boldsymbol{J}_{p_{iy}} e_{iy} \end{bmatrix} \tag{36} $$

$$ \boldsymbol{J}_{v_{2i}} = \frac{1}{d_i}\begin{bmatrix} {}^{i}\boldsymbol{J}_{p_{ix}}(h_{iz}+d_i) \\ {}^{i}\boldsymbol{J}_{p_{iy}}(h_{iz}+d_i) \\ -{}^{i}\boldsymbol{J}_{p_{ix}} h_{ix} - {}^{i}\boldsymbol{J}_{p_{iy}} h_{iy} + d_i {}^{i}\boldsymbol{J}_{p_{iz}} \end{bmatrix} \tag{37} $$

$\boldsymbol{J}_{{}^{i}\omega_p}$ is the Jacobian of the platform angular velocity stated in $i$th leg frame and is written as following:

$$ \boldsymbol{J}_{{}^{i}\omega_p} = \begin{bmatrix} 0_{3\times 3} & {}^{i}\boldsymbol{R}_{\boldsymbol{B}} \end{bmatrix} \tag{38} $$

### 3.4 Dynamics

The inverse dynamics problem is to determine the actuator forces for a given trajectory of the moving platform. Kane's equation of motion for multi-body systems can be expressed as (Makkar et al. 2005):

$$ \sum_{i=1}^{m}\left(\boldsymbol{f}_{\boldsymbol{i}}\cdot\frac{\partial \boldsymbol{v}_{\boldsymbol{i}}}{\partial \dot{x}_{pj}} + \boldsymbol{m}_{\boldsymbol{i}}\cdot\frac{\partial \boldsymbol{\omega}_{\boldsymbol{i}}}{\partial \dot{x}_{pj}}\right) = 0, \quad j = 1\ldots n \tag{39} $$

where $\boldsymbol{f}_{\boldsymbol{i}} = \check{\boldsymbol{f}}_{\boldsymbol{i}} + \check{\boldsymbol{f}}_{\boldsymbol{i}}^{*}$ and $\boldsymbol{m}_{\boldsymbol{i}} = \check{\boldsymbol{m}}_{\boldsymbol{i}} + \check{\boldsymbol{m}}_{\boldsymbol{i}}^{*}$. The $\boldsymbol{f}_{\boldsymbol{i}}$ and $\boldsymbol{m}_{\boldsymbol{i}}$ are the active force and moment applied on $i$th point of application on the manipulator by the external and inertial means. The $\boldsymbol{v}_{\boldsymbol{i}}$ and $\boldsymbol{\omega}_{\boldsymbol{i}}$ are linear and angular velocity of the $i$th point. The $\dot{x}_{pj}$ is $j$th element of the generalized velocity of the manipulator. Active forces and moments exerted on the center of mass of the moving platform are written as

$$ \begin{aligned} \boldsymbol{F}_p &= \begin{bmatrix} \boldsymbol{f}_{\boldsymbol{p}} \\ \boldsymbol{m}_{\boldsymbol{p}} \end{bmatrix} = \begin{bmatrix} \check{\boldsymbol{f}}_{\boldsymbol{p}} + \check{\boldsymbol{f}}_{\boldsymbol{p}}^{*} \\ \check{\boldsymbol{m}}_{\boldsymbol{p}} + \check{\boldsymbol{m}}_{\boldsymbol{p}}^{*} \end{bmatrix} \\ &= \begin{bmatrix} (\boldsymbol{f}_{\boldsymbol{e}} + m_p\boldsymbol{g}) + (-m_p\dot{\boldsymbol{v}}_{\boldsymbol{p}}) \\ (\boldsymbol{n}_{\boldsymbol{e}}) + \left(-{}^{B}\boldsymbol{I}_{\boldsymbol{p}}\dot{\boldsymbol{\omega}}_{\boldsymbol{p}} - \boldsymbol{\omega}_{\boldsymbol{p}} \times ({}^{B}\boldsymbol{I}_{\boldsymbol{p}}\boldsymbol{\omega}_{\boldsymbol{p}})\right) \end{bmatrix} \end{aligned} \tag{40} $$

where $\boldsymbol{f}_{\boldsymbol{e}}$ and $\boldsymbol{n}_{\boldsymbol{e}}$ are the external force and torque applied on the moving platform. The inertia matrix ${}^{B}\boldsymbol{I}_{\boldsymbol{p}} = {}^{B}\boldsymbol{R}_{\boldsymbol{P}}{}^{P}\boldsymbol{I}_{\boldsymbol{p}}{}^{P}\boldsymbol{R}_{\boldsymbol{B}}$ is the moving platform's moment of inertia about axes parallel to the fixed frame, with the origin located at the mass center of the platform.

The external forces acting on the center of masses of the leg links are only the gravitational force, and the active forces and moments for the legs will be

$$ \textit{First link}: \begin{cases} \boldsymbol{f}_{1i} = \check{\boldsymbol{f}}_{1i} + \check{\boldsymbol{f}}_{1i}^{*} = (m_e{}^{i}\boldsymbol{R}_{\boldsymbol{B}}\boldsymbol{g}) + (-m_e{}^{i}\dot{\boldsymbol{v}}_{1i}) \\ \boldsymbol{m}_{1i} = \check{\boldsymbol{m}}_{1i} + \check{\boldsymbol{m}}_{1i}^{*} = 0 + \left(-{}^{i}\boldsymbol{I}_{1i}{}^{i}\dot{\boldsymbol{\omega}}_{\boldsymbol{i}} - {}^{i}\boldsymbol{\omega}_{\boldsymbol{i}} \times ({}^{i}\boldsymbol{I}_{1i}{}^{i}\boldsymbol{\omega}_{\boldsymbol{i}})\right) \end{cases} \tag{41} $$

$$ \textit{Second link}: \begin{cases} \boldsymbol{f}_{2i} = \check{\boldsymbol{f}}_{2i} + \check{\boldsymbol{f}}_{2i}^{*} = (m_h{}^{i}\boldsymbol{R}_{\boldsymbol{B}}\boldsymbol{g}) + (-m_h{}^{i}\dot{\boldsymbol{v}}_{2i}) \\ \boldsymbol{m}_{2i} = \check{\boldsymbol{m}}_{2i} + \check{\boldsymbol{m}}_{2i}^{*} = 0 + \left(-{}^{i}\boldsymbol{I}_{2i}{}^{i}\dot{\boldsymbol{\omega}}_{\boldsymbol{i}} - {}^{i}\boldsymbol{\omega}_{\boldsymbol{i}} \times ({}^{i}\boldsymbol{I}_{2i}{}^{i}\boldsymbol{\omega}_{\boldsymbol{i}})\right) \end{cases} \tag{42} $$

where ${}^{i}\boldsymbol{I}_{1i}$ and ${}^{i}\boldsymbol{I}_{2i}$ are the leg's lower and upper link moments of inertia about axes parallel to the leg local frame $C_i(x_i, y_i, z_i)$, passing through their corresponding centers of masses.

Spherical and universal joints are only affected by the friction in the form of an active moment as

$$ \textit{Spherical joints}: \begin{cases} \boldsymbol{f}_{sp_i} = 0 \\ \boldsymbol{m}_{sp_i} = \check{\boldsymbol{m}}_{sp_i} + \check{\boldsymbol{m}}_{sp_i}^{*} = \left(-\boldsymbol{fm}_{sp_i}\right) + (0) \end{cases} \tag{43} $$

$$ \textit{universal joints}: \begin{cases} \boldsymbol{f}_{un_i} = 0 \\ \boldsymbol{m}_{un_i} = \check{\boldsymbol{m}}_{un_i} + \check{\boldsymbol{m}}_{un_i}^{*} = (-\boldsymbol{fm}_{un_i}) + (0) \end{cases} \tag{44} $$

where $\boldsymbol{fm}_{spi}$ and $\boldsymbol{fm}_{uni}$ are the friction moments for the $i$th spherical and universal joint.

These vectors are functions of $\left({}^i\boldsymbol{\omega}_p - {}^i\boldsymbol{\omega}_i\right)$ and ${}^i\boldsymbol{\omega}_i$.

Using Eqs. (39)–(44), the equation of motion can be obtained as

$$\boldsymbol{F}_p \cdot \frac{\partial \underline{\boldsymbol{x}}_p}{\partial \underline{\boldsymbol{x}}_{pj}} + \sum_{i=1}^{6} \Bigg( \boldsymbol{f}_{1i} \cdot \frac{\partial {}^i\boldsymbol{v}_{1i}}{\partial \underline{\boldsymbol{x}}_{pj}} + \boldsymbol{m}_{1i} \cdot \frac{\partial {}^i\boldsymbol{\omega}_i}{\partial \underline{\boldsymbol{x}}_{pj}} + \boldsymbol{f}_{2i} \cdot \frac{\partial {}^i\boldsymbol{v}_{2i}}{\partial \underline{\boldsymbol{x}}_{pj}} + \boldsymbol{m}_{2i} \cdot \frac{\partial {}^i\boldsymbol{\omega}_i}{\partial \underline{\boldsymbol{x}}_{pj}} + \boldsymbol{m}_{uni} \cdot \\ \times \frac{\partial {}^i\boldsymbol{\omega}_i}{\partial \dot{x}_{pj}} + \boldsymbol{m}_{spi} \cdot \frac{\partial \left({}^i\boldsymbol{\omega}_p - {}^i\boldsymbol{\omega}_i\right)}{\partial \underline{\boldsymbol{x}}_{pj}} \Bigg) + \left(\boldsymbol{\tau} - \boldsymbol{F}_{pr}\right) \cdot \frac{\partial \dot{\boldsymbol{q}}}{\partial \dot{x}_{pj}} = 0,\ j = 1 \ldots 6 \tag{45}$$

where $\boldsymbol{\tau}_{6\times 1}$ is the actuator's force vector and

$$\boldsymbol{F}_{pr} = \begin{bmatrix} \boldsymbol{f}_{pr1} & \cdots & \boldsymbol{f}_{pr6} \end{bmatrix}^{\mathrm{T}}$$

denotes the friction in the prismatic joints and is function of $\begin{bmatrix} \dot{d}_1 & \cdots & \dot{d}_6 \end{bmatrix}$. It is noted that $\boldsymbol{F}_p \cdot \frac{\partial \dot{\boldsymbol{x}}_p}{\partial \dot{x}_{pj}}$ is equal to $\boldsymbol{f}_p \cdot \frac{\partial \dot{\boldsymbol{t}}_p}{\partial \dot{x}_{pj}} + \boldsymbol{m}_p \cdot \frac{\partial \boldsymbol{\omega}_p}{\partial \dot{x}_{pj}}$. Since the derivatives of Eq. (45) are the different columns of the Jacobian matrices, this equation can be written in matrix form as the following:

$$\boldsymbol{F}_p + \sum_{i=1}^{6} \left( \boldsymbol{J}_{v_{1i}}^T \boldsymbol{f}_{1i} + \boldsymbol{J}_{v_{2i}}^T \boldsymbol{f}_{2i} + \boldsymbol{J}_{{}^i\omega_i}^T \left(\boldsymbol{m}_{1i} + \boldsymbol{m}_{2i} + \boldsymbol{m}_{uni} - \boldsymbol{m}_{spi}\right) + \boldsymbol{J}_{{}^i\omega_p}^T \boldsymbol{m}_{spi} \right) \\ + \boldsymbol{J}_q^T \left(\boldsymbol{\tau} - \boldsymbol{F}_{pr}\right) = 0 \tag{46}$$

where $\frac{\partial {}^i\boldsymbol{v}_{1i}}{\partial \dot{x}_{pj}}, \frac{\partial {}^i\boldsymbol{v}_{2i}}{\partial \dot{x}_{pj}}, \frac{\partial {}^i\boldsymbol{\omega}_i}{\partial \dot{x}_{pj}}, \frac{\partial \left({}^i\boldsymbol{\omega}_p - {}^i\boldsymbol{\omega}_i\right)}{\partial \dot{x}_{pj}}$, and $\frac{\partial \dot{\boldsymbol{q}}}{\partial \dot{x}_{pj}}$ are the $j$th column of $\boldsymbol{J}_{v_{1i}}, \boldsymbol{J}_{v_{2i}}, \boldsymbol{J}_{{}^i\omega_i}, \boldsymbol{J}_{{}^i\omega_p} - \boldsymbol{J}_{{}^i\omega_i}$, and $\boldsymbol{J}_q$. It is obvious that $\frac{\partial \dot{\boldsymbol{x}}_p}{\partial \dot{x}_{pj}}$ is the $j$th column of the $6 \times 6$ identity matrix. Equation (46) describes the dynamics of the general Stewart platform manipulator, and this equation can be further simplified to the following equation:

$$\boldsymbol{F}_p + \boldsymbol{J}_x^T \boldsymbol{F}_x + \boldsymbol{J}_y^T \boldsymbol{F}_z + \boldsymbol{J}_q^T \left(\boldsymbol{F}_z + \boldsymbol{\tau} - \boldsymbol{F}_{pr}\right) + \sum_{i=1}^{6} \boldsymbol{M}_{sp_i} = 0 \tag{47}$$

where

$$\boldsymbol{J}_x = \begin{bmatrix} {}^1\boldsymbol{J}_{p_{1x}} \\ \vdots \\ {}^6\boldsymbol{J}_{p_{6x}} \end{bmatrix} \quad \boldsymbol{J}_y = \begin{bmatrix} {}^1\boldsymbol{J}_{p_{1y}} \\ \vdots \\ {}^6\boldsymbol{J}_{p_{6y}} \end{bmatrix} \quad \boldsymbol{M}_{sp_i} = \begin{bmatrix} 0_{3\times 1} \\ {}^B\boldsymbol{R}_i \boldsymbol{m}_{sp_i} \end{bmatrix} \quad \boldsymbol{F}_z = \begin{bmatrix} f21_z \\ f22_z \\ \vdots \\ f26_z \end{bmatrix}$$

$$\mathbf{F_x} = \begin{bmatrix} \left(e_{1_z} f11_x - e_{1_x} f11_z + \left(h_{1_z} + d_1\right) f21_x - h_{1_x} f21_z + m11_y + m21_y + m_{un1_y} - m_{sp1_y}\right)/d_1 \\ \left(e_{2_z} f12_x - e_{2_x} f12_z + \left(h_{2_z} + d_2\right) f22_x - h_{2_x} f22_z + m12_y + m22_y + m_{un2_y} - m_{sp2_y}\right)/d_2 \\ \vdots \\ \left(e_{6_z} f16_x - e_{6_x} f16_z + \left(h_{6_z} + d_6\right) f26_x - h_{6_x} f26_z + m16_y + m26_y + m_{un6_y} - m_{sp6_y}\right)/d_6 \end{bmatrix}$$

$$\mathbf{F_y} = \begin{bmatrix} \left(e_{1_z} f11_y - e_{1_y} f11_z + \left(h_{1_z} + d_1\right) f21_y - h_{1_y} f21_z - m11_x - m21_x - m_{un1x} + m_{sp1x}\right)/d_1 \\ \left(e_{2_z} f12_z - e_{2_y} f12_z + \left(h_{2_z} + d_2\right) f22_y - h_{2_y} f22_z - m12_x - m22_x - m_{un2_x} + m_{sp2_x}\right)/d_2 \\ \vdots \\ \left(e_{6_z} f16_z - e_{6_y} f16_z + \left(h_{6_z} + d_6\right) f26_y - h_{6_y} f26_z - m16_x - m26_x - m_{un6_x} + m_{sp6_x}\right)/d_6 \end{bmatrix}$$

where $x$, $y$, and $z$ indexes indicate the related components of their vectors (Fig. 2).

Finally, the inverse dynamic equation is written as

$$\boldsymbol{\tau} = \boldsymbol{F}_{pr} - \boldsymbol{F}_z - \boldsymbol{J}_q^{-T} \left( \boldsymbol{F}_p + \boldsymbol{J}_x^T \boldsymbol{F}_x + \boldsymbol{J}_y^T \boldsymbol{F}_z + \sum_{i=1}^{6} \boldsymbol{M}_{sp_i} \right) \tag{48}$$

### 3.5 Friction

Here, friction is modeled generally in the equation of motion. Among different models in the literature, the one developed by Makkar et al. (2005) is more suitable for control purposes since it is a continuous function, whose derivative exists over its domain and covers the important properties of the friction force. This model includes static, coulomb, and viscous friction, and it also covers the Stribeck effect (Fig. 3). Efficiency of the model is proved and it has the following parametric form:

$$F_{fric}(u) = c_1\left(\tanh(c_2 u) - \tanh(c_3 u)\right) + c_4 \tanh(c_5 u) + c_6 u \tag{49}$$

The parameter $\mu$ denotes the relative speed between the two parts. The term $(C_1 + C_4)$ denotes the static friction, and the term (tanh $(c_2 u)$ − tanh $(c_3 u)$) illustrates the Stribeck effect. The Coulomb friction is denoted by $c_4$ tanh $(c_5 u)$, and the last term is related to the viscous friction. Further information is provided in Ref. [12].

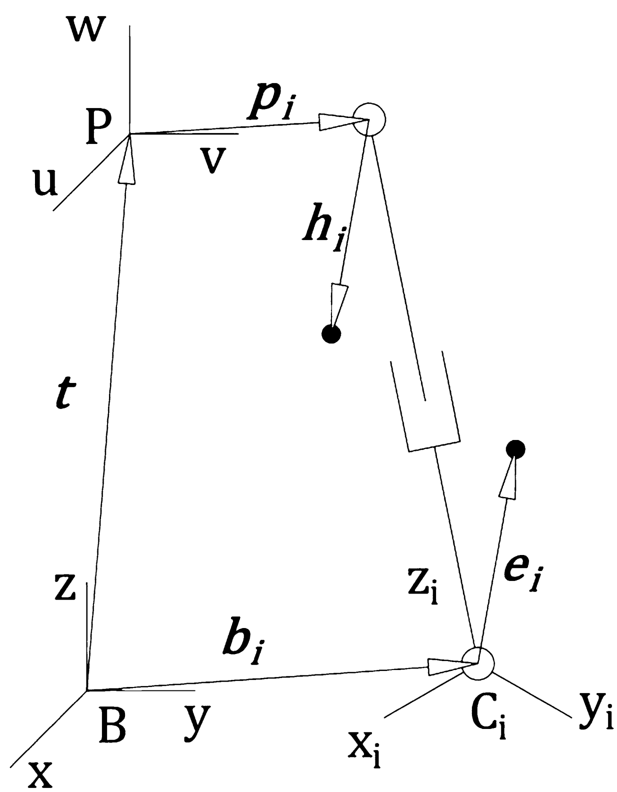


**Fig. 2** Details of a typical leg

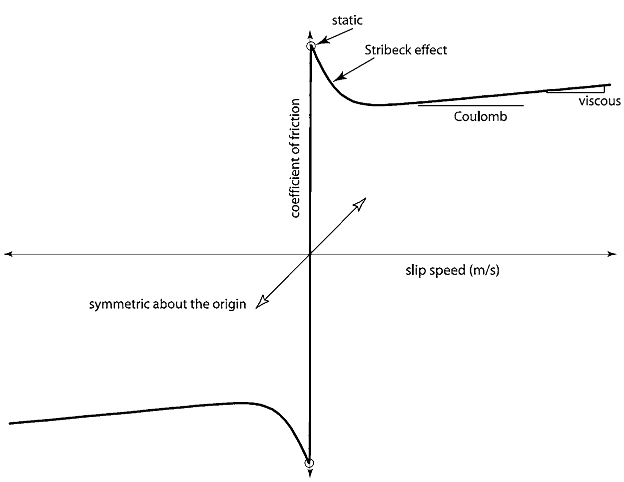


**Fig. 3** Friction different aspects [12]

### 3.6 Simulations

A fifth-order polynomial is fitted between points $x_{initial} = [-0.15 \quad 0.15 \quad 0.8 \quad -10 \quad 0 \quad 0]^T$ and $x_{final} = [0.15 \quad -0.15 \quad 0.8 \quad 0 \quad 10 \quad 0]^T$ so that the initial and final velocities and accelerations have zero values, the orientation is stated in degrees. An external force of (20 sin $t$)($N$) along the $z$ axis of the fixed frame, and an external torque of

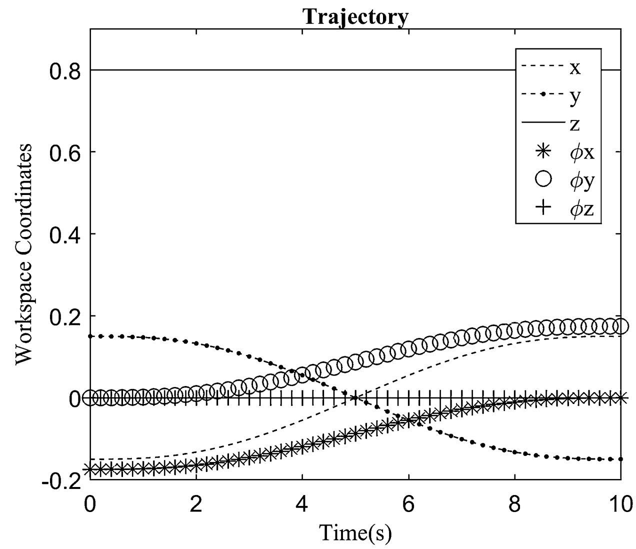


**Fig. 4** Workspace trajectory

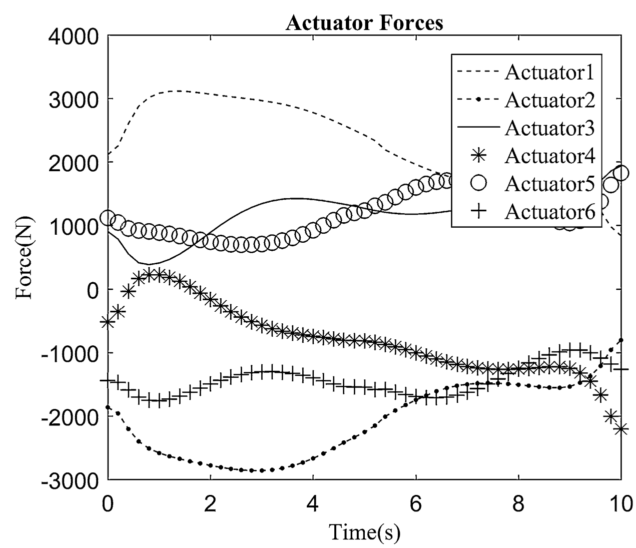


**Fig. 5** Leg forces

(25 sin $t$)($N$ $m$) along the $x$ axis of the fixed frame are exerted on the center of mass of the moving platform. Simulation results are shown in Figs. 4, 5, 6, and 7.

Results from Fig. 6 show that friction has a great effect on the leg forces. Figure 7 also shows that simplifying geometrical and dynamical properties of legs affect the leg forces, though not as great as the friction effect, but neglecting it produces difficulties in fine controlling the manipulator.

### 3.7 Computational Cost

The computational cost of the algorithm is evaluated by the MAPLE®. This is also done for the inverse dynamic model based on virtual work principle (Tsai 2000) by Tsai and

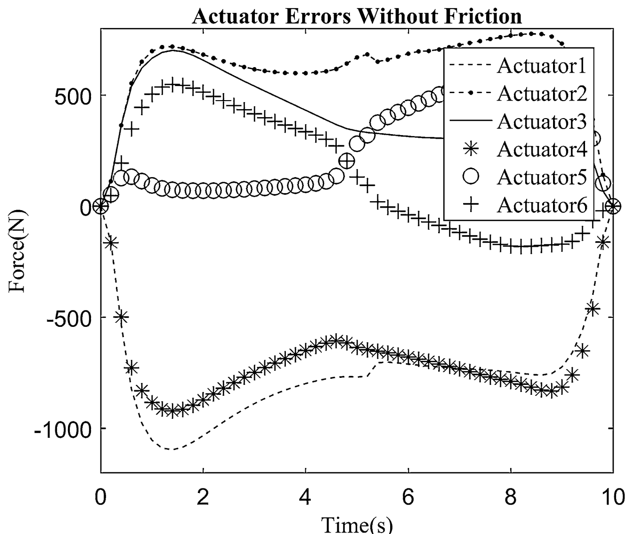


**Fig. 6** Actuator forces errors without friction

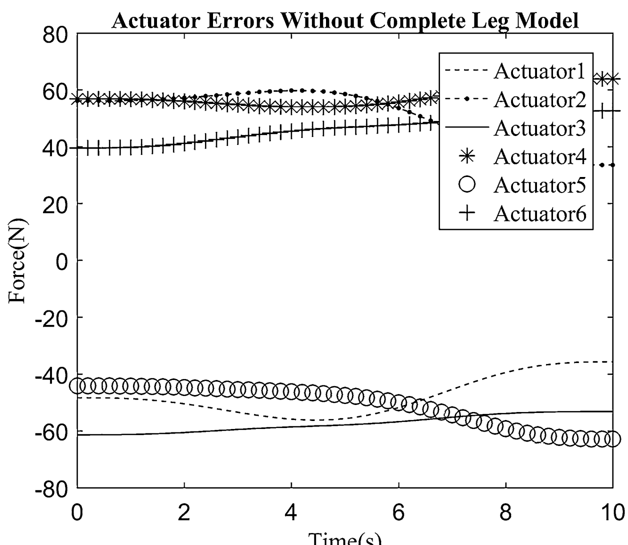


**Fig. 7** Actuator forces error by neglecting legs center of masses vector

Newton–Euler-based model (Dasgupta and Mruthyunjaya 1998) by Dasgupta. It must be noted that friction is ignored in the calculations to make the results comparable, and in the Newton–Euler (Dasgupta and Mruthyunjaya 1998) case only equations which are needed for the inverse dynamics have been considered in the computational cost determination. Results are presented in Table 1. However, computational cost of the proposed method is slightly higher than that of the method presented in (Tsai 2000), but the proposed method does not contain the leg simplifications made in Tsai's (2000) work. It is obvious that the presented algorithm is computationally lighter than the Newton–Euler method (Dasgupta and Mruthyunjaya 1998) with the same dynamic and geometric properties in its formulation.

**Table 1** Computational cost

| Operation | Virtual work principle (Tsai 2000) | Newton–Euler method | Kane method |
|---|---|---|---|
| Multiply/division | 1430 | 2244 | 1688 |
| Addition | 955 | 1712 | 1075 |
| Trigonometric | 24 | 24 | 24 |
| Square root | 12 | 12 | 12 |
| $6 \times 6$ matrix inversion | 1 | 1 | 1 |

## 4 Conclusion

The inverse dynamics problem of the Stewart platform manipulator is solved using the Kane's method. The presented algorithm is straightforward with no simplification made on the manipulator properties. Friction is modeled in a general form of a function, hence creating no specific limitations on the form of the friction used. The effect of neglecting friction and simplifying leg models is also presented. Although no important parameter is omitted in modeling of the manipulator's properties, the algorithm is fast and is suitable for real-time applications.

## Appendix

System parameters:

All parameters are in SI units.

Universal joint locations:

$$[\boldsymbol{b}_1 \quad \cdots \quad \boldsymbol{b}_6] = \begin{bmatrix} 0.483 & 0.354 & -0.354 & -0.483 & -0.129 & 0.129 \\ 0.129 & 0.354 & 0.354 & 0.129 & -0.483 & -0.482 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Spherical joint locations:

$$[\boldsymbol{p}'_1 \quad \cdots \quad \boldsymbol{p}'_6] = \begin{bmatrix} 0.068 & -0.068 & -0.197 & -0.129 & 0.129 & 0.197 \\ 0.188 & 0.188 & -0.035 & -0.153 & -0.153 & -0.035 \\ -0.05 & -0.05 & -0.05 & -0.05 & -0.05 & -0.05 \end{bmatrix}$$

Mass of each part:

$m_p = 30$

$m_e = 2$

$m_h = 1.2$

Legs' center of masses:

$\boldsymbol{e} = [0.05 \quad 0.03 \quad 0.25]^{\mathrm{T}}$

$\boldsymbol{h} = [0.03 \quad 0.03 \quad -0.2]^{\mathrm{T}}$

Moments of inertia of each part:

$$ {}^{p}\boldsymbol{I}_{p} = \begin{bmatrix} 2.5 & 0 & 0 \\ 0 & 2.5 & 0 \\ 0 & 0 & 5 \end{bmatrix} $$

$$ {}^{i}\boldsymbol{I}_{1i} = \begin{bmatrix} 0.11 & 0.15 & 0.4 \\ 0.15 & 0.2 & 0.3 \\ 0.4 & 0.3 & 0.5 \end{bmatrix} $$

$$ {}^{i}\boldsymbol{I}_{2i} = \begin{bmatrix} 0.17 & 0.07 & 0.15 \\ 0.07 & 0.15 & 0.13 \\ 0.15 & 0.13 & 0.25 \end{bmatrix} $$

Friction coefficients of the joints:
Prismatic joints (actuators):

$$ [\, c_1 \quad \cdots \quad c_6 \,] = [\, 20 \quad 120 \quad 95 \quad 10 \quad 1150 \quad 0.65 \,] $$

Universal joints:

$$ [\, c_1 \quad \cdots \quad c_6 \,] = [\, 1 \quad 150 \quad 80 \quad 2 \quad 1000 \quad 0.05 \,] $$

Spherical joints:

$$ [\, c_1 \quad \cdots \quad c_6 \,] = [\, 5 \quad 120 \quad 80 \quad 3 \quad 860 \quad 0.15 \,] $$

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